

Supplemental Material

Evidence for quasicritical brain dynamics

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SIMULATIONS AND NETWORKS

Simulations of the Cortical Branching Model (CBM) were carried out on random, strongly-connected directed networks. That is, whereas loops are not allowed, every node in the network is accessible from any other node in the network. An example is presented in Fig. 1.



Figure 1. Example of a $L = 128$ random, strongly-connected directed network used in our CBM simulations.

For each set of fixed parameter values (L , κ , p_s , k_{in} , and τ_r), a sample of ten different randomly-generated networks is used to simulate a total of 10^5 avalanches. In order to find the optimal avalanche distributions, the peak of the dynamical susceptibility $\chi(\kappa)$ is found by sweeping across κ values while leaving all other parameter values fixed. Avalanche distributions at the peak κ value (the Widom κ_w), and also those within the quasicritical scaling regime [7], display scale-free behavior over the largest number of decades.

SELECTION CRITERIA OF EXPERIMENTAL DATA

A method similar to the one used in [11] was initially used to analyze experimental data. How-

ever, this method reduced the statistical sample of avalanches beyond a reasonable point and we hence designed a new analysis method.

Data rasters from each sample were first partitioned into windows of duration $\Delta T = 10$ s. Within each window, the branching ratio σ , dynamical susceptibility χ , and local time fluctuation LTF were calculated (see Fig. 2, left column). Peak values of χ were then associated with values of σ and the LTF (see Fig. 2, right column). The values of the LTF corresponding to the peak χ fell within a standard deviation of the mean LTF, $\langle \text{LTF} \rangle$, or the flattest part of the sorted LTF. The sorted LTF is obtained by sorting the values of the LTF in ascending order (see Fig. 3). This flat region can be found to the right of the peak of the corresponding long-tailed probability distribution. Thus, by selecting only those avalanches found within windows corresponding to this flat LTF region, we restrict our analysis to those avalanches within the quasicritical regime at and around the peak χ .

Consequently, avalanches were selected from the entire 1-hr recordings of data sets which satisfied the following conditions:

- A large statistical sample of avalanches, $\mathcal{O}(10^4)$.
- An accumulation of LTF values around the same σ where the susceptibility is maximal.
- Minimal fluctuations away from the mean LTF, $\langle \text{LTF} \rangle$.
- A clear inflexion (or “bump”) in the avalanche distributions due to finite-size effects at the peak of maximum susceptibility.
- Agreement with the scaling relation $\gamma = (1 - \tau_T)/(1 - \tau_S)$.

Data sets which satisfy all of these conditions feature $\langle \text{LTF} \rangle = 2.06 \pm 0.07$ (see Fig. 3). Consistency with the γ scaling relation can generally be met without invoking the other conditions. The time-binning of avalanche samples can be adjusted to modify the shape of the avalanche distributions, as we describe in the following section.

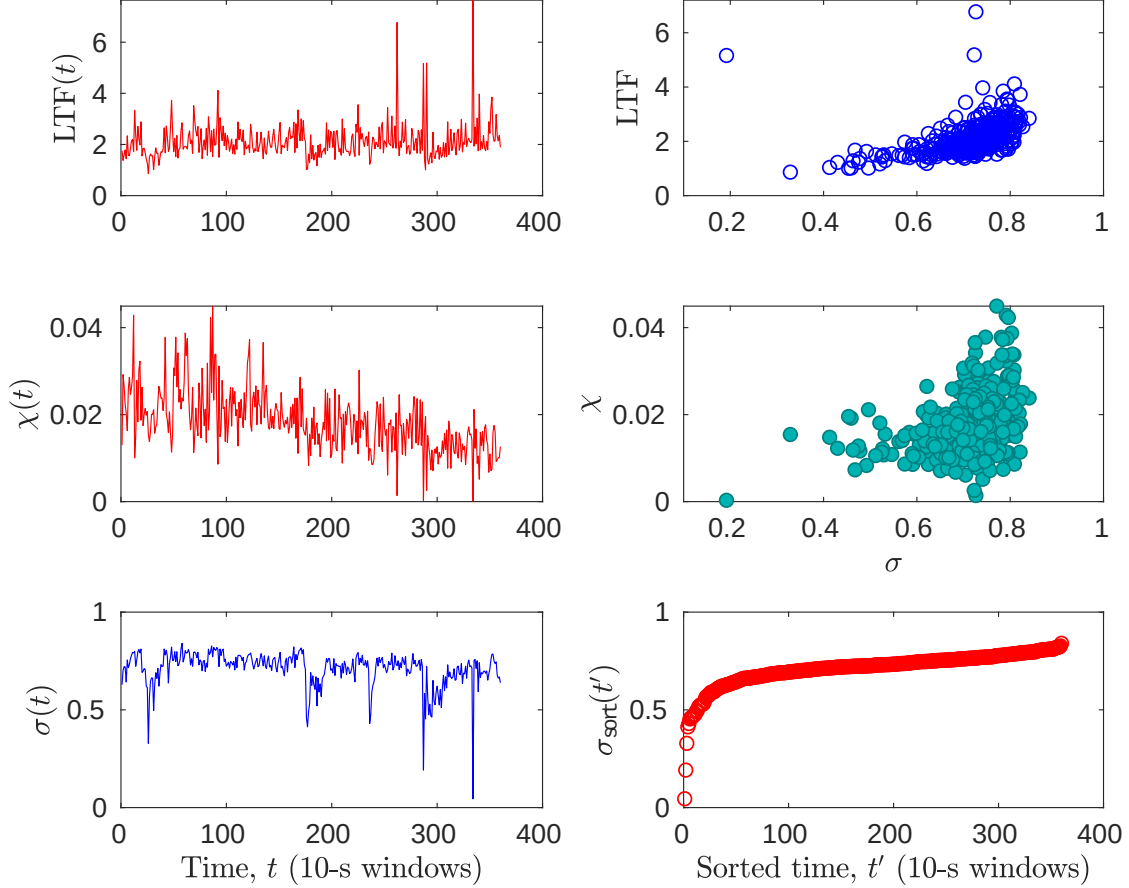


Figure 2. Analysis for mouse data set 18. The left column shows plots for $LTF(t)$, $\chi(t)$, and $\sigma(t)$, where time t is in units of 10-s windows. Scatter plots in the right column illustrate how the peak values of $\chi(\sigma)$ correspond to certain values of $LTF(\sigma)$. Indeed, the flattest portions of the sorted σ and LTF values ($\sigma_{\text{sort}}(t')$ and $LTF_{\text{sort}}(t')$, see Fig. 3) correspond to the peak value of the dynamical susceptibility (roughly falling between $\sigma = 0.6$ and 0.8).

EFFECT OF DATA BINNING AND FINITE SIZE ON EFFECTIVE EXPONENTS

We selected a data binning $\delta t \in [1 \text{ ms}, 10 \text{ ms}]$ which gives a clear power-law distribution, provides enough avalanches for statistical analyses, and also satisfies the criteria stated in the previous section. For example, see Fig. 5, which shows avalanche distributions for mouse data set 4 binned at $\delta t = 2 \text{ ms}$ and $\delta t = 6 \text{ ms}$ (at $\delta t = 1 \text{ ms}$, this data set remains subcritical). Both distributions show a clear inflexion, indicating proximity to peak susceptibility. However, at $\delta t = 2 \text{ ms}$, a few issues violate our criteria: (1) The $\langle LTF \rangle$ is much larger than the other data sets, (2) There are not many points on which to fit the exponent line, resulting in a large uncertainty, and (3) A clear peak in χ is not present.

On the other hand, when a 6-ms binning is used, the mean value for mouse data set 4 falls within a

standard deviation of the expected value $\langle LTF \rangle$. The effective exponents in Fig. 5, $\tilde{\tau}_S$ and $\tilde{\tau}_T$, can be seen to rise as a result of this adjustment in bin size.

Most of the mouse data showed to be either supercritical or subcritical when binned at $\delta t = 1 \text{ ms}$. However, even though clear peaks in the susceptibility were not found even after changing the binning, their effective critical exponents still follow the scaling line (see Fig. 4).

Other extrinsic factors also influence values of the measured effective exponents. We ran simulations of the CBM at $p_s = 10^{-3}$, performing a finite-size scaling along a roughly determined Widom line, calculating the effective exponents along the way (values in Table I). Our results show that while the different system sizes agree with the γ scaling law, the exponent values decrease as the system size is increased, giving in the thermodynamic limit approximately 1.81 and 2.18 for the size and duration ex-

ponents, respectively.

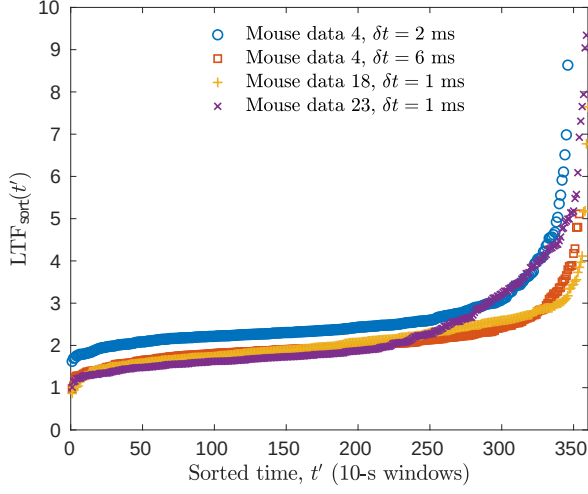


Figure 3. Effect of time binning on the sorted LTF. Adjusting the time binning from $\delta t = 2$ ms to $\delta t = 6$ ms lowers the mean LTF for mouse data set 4 to within one standard deviation of $\langle \text{LTF} \rangle = 2.06 \pm 0.07$, the mean for data sets which satisfy all of our strict criteria.

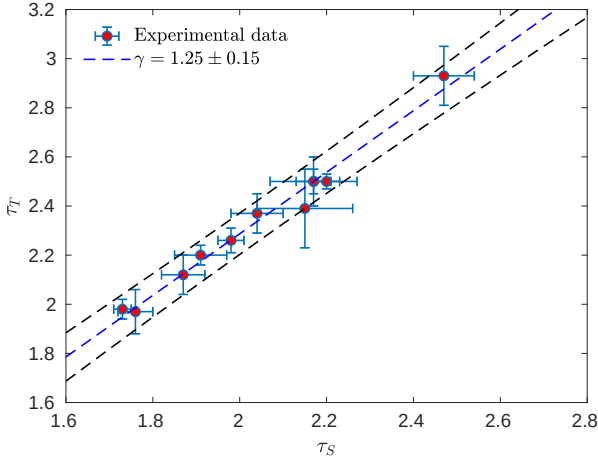


Figure 4. Scaling line for mouse data sets with 1-ms binning. Many mouse data sets which fail to satisfy all of our strict selection criteria, still satisfy the γ scaling line.

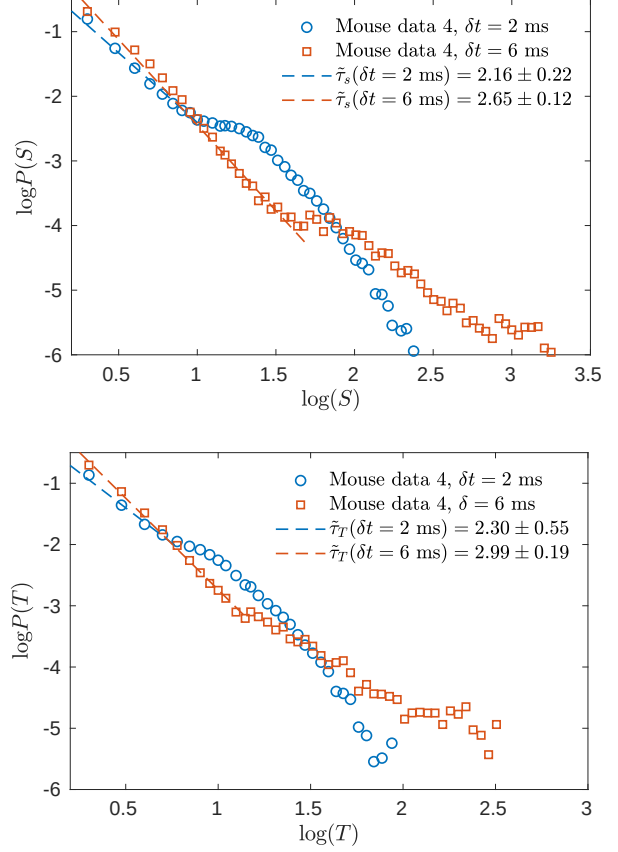


Figure 5. Effective critical exponents for mouse data 4 binned at 2ms. Effective critical exponents for mouse data 4 binned at 6ms.

L	κ_w	$\tilde{\tau}_S$	$\tilde{\tau}_T$	$\frac{\tilde{\tau}_T - 1}{\tilde{\tau}_S - 1}$	γ
64	1.16	1.65(2)	1.96(6)	1.47(5)	1.50(7)
128	1.10	1.57(1)	1.82(1)	1.44(3)	1.51(5)
256	1.07	1.45(1)	1.59(3)	1.31(7)	1.41(2)

Table I. Effective critical exponents for avalanche size and duration distributions at $p_s = 10^{-3}$ with different network sizes L .